JAN–D–56A

16 MARCH 1945

SUPERSEDING
JAN–D–56
30 June 1944

JOINT ARMY-NAVY SPECIFICATION

DICHLORODIPHENYL TRICHLOROETHANE (DDT)

Army Number
4-1122A

Navy Number
51D14a

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
4–1122	51D14
30 June 1944	30 June 1944

A. APPLICABLE SPECIFICATIONS.

A–1. The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–P–107—Packaging and Packing for Overseas Shipment—Boxes, Wood, Wire-bound.
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–124—Packaging and Packing for Overseas Shipment—Cans and Pails, Metal.

U. S. ARMY SPECIFICATIONS

3–181—Enamel, Olive-Drab, Rust-Inhibiting.
100–2—Standard Specification for Marking Shipments by Contractors.

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[1]
52P63—Paraffin, Chlorinated (Resinous).

B. TYPE AND GRADES.

B–1. DDT covered by this specification shall be of but one type and the following grades as specified in the contract or order. (See par. H–2a):

Grade A—DDT, Aerosol.
Grade B—DDT, Technical.

[1] Applicable only to Navy purchases.

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C. MATERIAL AND WORKMANSHIP.

C–1. Each grade shall comprise 2, 2-bis-(p-chlorophenyl)-1, 1, 1-trichloroethane, free from added modifying agents, and shall comply with the detail requirements specified for the applicable grade.

D. GENERAL REQUIREMENTS.

D–1. See section E.

E. DETAIL REQUIREMENTS.

E–1. *Chemical and physical properties.*—The chemical and physical properties shall conform to the requirements specified in table I.

TABLE I.—*Chemical and physical properties*

Property	Grade A		Grade B	
	Minimum	Maximum	Minimum	Maximum
Setting point, °C			90	
Melting point, °C	103			
Organic chlorine, percent by weight	49.5	50.5	48	51
Ash content, percent by weight		0.05		0.5
Water insoluble, steam volatile material		(1)		(2)
Chloral hydrate, percent by weight		.025		.025
pH by distillation			4.5	8.0
pH by extraction	5.5	7.5	5.0	8.0
Water soluble material, percent by weight		.05		.25
Cyclohexanone insoluble, ml				.2
Monofluorotrichloromethane insoluble, ml		.01		

[1] 0.1 ml/50 gm.
[2] 0.1 ml/20 gm.

E–2. *Form.*—

E–2a. *Grade A—aerosol.*—Grade A shall be a fine white crystalline powder, free flowing and without lumps.

E–2b. *Grade B—technical.*—Grade B shall be a fine to medium granular powder with a white to cream color.

E–3. *Instruction marking (for Navy purchases of Grade B only).*—Containers shall be marked with the following:

POISON.—Use only under the direction of a medical officer. DDT is poisonous when taken by mouth; it should not be stored in galleys or other places where it may be confused with flour or other powdered foods. Oil solutions of DDT are absorbed through the skin; occasional contact with the skin is apparently not dangerous, but prolonged or repeated contact must be avoided.

Insecticide concentrate powder (DDT), Dichloro-diphenyl-trichloroethane, Joint Army-Navy Specification JAN–D–56.

This is a concentrate to be used as an ingredient in insecticides of the following formulations:

As an aqueous emulsion for louseproofing garments or for a residual type spray

Make a concentrated stock solution by mixing 5 parts by weight of this powder (DDT) with 11 parts by weight of xylene and 4 parts by weight of a suitable synthetic-wetting agent. Add this stock solution slowly to water with continuous mixing to

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form an aqueous emulsion. The proportions should be approximately 7 fluid ounces to each gallon of water for treating garments, and approximately 1 quart to each gallon of water when used as a residual type spray.

As a petroleum oil solution for use to control breeding of mosquito larvae

Dissolve this powder (DDT) in petroleum oil in the approximate proportions of 7 ounces to each gallon of oil. Stir at intervals until all of the powder is dissolved.

As a liquid insecticide for use in standard insecticide sprayers

Dissolve this powder (DDT) in crude kerosene in the approximate proportions of 7 ounces to each gallon of kerosene. Stir at intervals until all of the powder is dissolved. If refined kerosene is used it may be necessary to add up to 5 percent of cyclohexanone or xylene in order to dissolve all of the powder. It may be desirable to filter out particles of foreign matter before use.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS.

F–1. *Sampling.*—

F–1a. *Lot.*—For purposes of sampling, a lot shall consist of a numbered batch, or of a manufacturer's numbered blended control lot.

F–1b. *Sampling procedure for lot acceptance tests.*—The Government inspector shall select a suitable sample from each lot for the tests specified in paragraph F–3a.

F–1c. *Sampling procedure for production check tests.*—From the first lot offered for delivery under a contract or order, and from every 60 lots thereafter, or at intervals of 30 days, whichever is the more frequent, the Government inspector shall take at random a 1-pound sample, which shall be sent to a Government laboratory designated by the bureau or agency concerned for the tests specified in paragraph F–3b (1).

F–2. *Inspection.*—Inspection to determine compliance with the requirements of this specification shall be made by the Government inspector at the place of manufacture.

F–3. *Tests.*—

F–3a. *Lot acceptance tests (at place of manufacture).*—The sample selected as specified in paragraph F–1b shall be subjected to the following tests:

Setting point. (See par. F–4a.)
Melting point. (See par. F–4b.)
Ash content. (See par. F–4d.)
Determination of pH by extraction. (See par. F–4e (2).)
Monofluorotrichloromethane insoluble material. (See par. F–4j.)

If the sample fails any of these tests, the lot shall be rejected.

F–3b. *Production check tests (at a Government laboratory).*—

F–3b (1). *Applicable tests.*—The sample selected as specified in paragraph F–1c shall be subjected to all the tests described in paragraph F–4.

F–3b (2). *Action in case of failure.*—The first lot offered for delivery under a contract or order shall not be accepted until a satisfactory test report is received on the production check test sample.

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(See par. F–1c.) Thereafter, except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling, inspection, and tests of paragraphs F–1b, F–2, and F–3a, and acceptance shall not be withheld pending receipt of test reports or production check test samples. However, upon receipt of an unsatisfactory test report on a production check test sample, the Government inspector shall select additional samples from every subsequent lot offered for delivery. The samples so selected shall be submitted to the Government laboratory, and shall there be subjected to the test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis of paragraphs F–1b, F–2, and F–3a when four successive lots have been accepted.

F–4. *Test details.*—

F–4a. *Setting point.*—Transfer 30±0.1 gm. of sample into a 40-ml. centrifuge tube (round bottom, pour-out, heavy duty—Corning No. 8260, or equivalent). The tube and its contents shall be supported in an oil bath maintained at 115°–120° C. When the DDT has almost all become molten, a glass ring stirrer and an accurate thermometer graduated in fifths of a degree (A. S. T. M. standard thermometer 70°–160° C.—Fisher Scientific Co. No. 15–168–R–3, or equivalent) shall be fitted into the tube through a two-hole cork stopper. While the DDT is being heated, a round, wide-mouth, 8-ounce, glass sample jar (approximately 2 inches in diameter and 5 inches high) shall be clamped upright in a water bath maintained at 70°±2° C. The jar shall be kept approximately 80 percent immersed throughout the entire determination and its mouth shall be fitted with a cork stopper into which one hole (just large enough to admit the tube containing the molten DDT) has been bored. When the DDT has melted completely and has reached a temperature of 115°–120° C., transfer the test tube and its contents from the oil bath to the water bath—sample jar system, fitting the length of the tube down into the jar through the one-holed cork stopper so that the top rim of the tube just protrudes over the upper surface of the cork. The liquefied sample shall now be stirred continuously, using the glass ring stirrer until the point of maximum supercooling has been reached and the temperature has begun to rise. Thereafter, the stirring shall be by means of the thermometer. All stirring shall be at the approximate rate of 100 strokes per minute (1 up and 1 down=1 stroke), all strokes to be 3–4 cm. in length, without breaking the upper surface of the sample. The liquefied material shall be stirred while it supercools. When the temperature drops to 88° C. a small amount of the sample of DDT being tested may be added as seed crystals if none have already formed in the test tube. After the material begins to crystallize, the temperature rises. At this point the stirring shall be stopped momentarily every 15 seconds and a temperature reading taken. The highest temperature reached after the point of maximum supercooling shall be taken as the setting point. The test shall be discontinued after two successively lower readings after the point of maximum supercooling.

F–4b. *Melting point.*—Melting point shall be determined by the capillary tube method, using a Thiele-Dennis, or equivalent, type melting tube, heated at a rate not to exceed 0.5° C. per minute. The temperature at which the sample becomes completely clear shall be the melting point.

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F–4c. *Organically bound chlorine.*—The modified Winter method specified in paragraph F–4c (1) is recommended. The use of the Parr peroxide bomb method described in paragraph F–4c (2) will be permitted. However, in the event of dispute the results obtained with the modified Winter method shall govern. The correction factor applicable to the method used shall be determined using the following equation:

$$f=\frac{T}{A}$$

where

f=correction factor.
T=theoretical calculated value for chlorine in pure, recrystallized DDT.
A=apparent value for chlorine in pure, recrystallized DDT as determined by the method used.

The results obtained for each sample by a given method shall be multiplied by the applicable correction factor.

F–4c (1). *Winter method.*—The chlorine content of the DDT shall be determined using the apparatus illustrated by figure 1.

F–4c (1) a. *U-tube.*—The U-tube for weighing and volatizing the sample of DDT shall be made of 9-mm. glass tubing on the longer side and 1-mm. glass capillary tubing on the shorter tide as illustrated by figure 1.

F–4c (1) b. *Heater device.*—The device for heating and volatizing the sample of DDT in the U-tube shall consist of approximately 10 feet of No. 28 American wire gage (0.01264-inch diameter) nichrome wire, wound 10 turns per inch over a strip of narrow asbestos tape which shall be placed over the section of the tube indicated on figure 1. The heater shall be controlled by a suitable variable auto-transformer.

F–4c (1) c. *Chimney.*—The chimney shall be a glass adapter having an 18-mm. inside diameter at its larger opening.

F–4c (1) d. *Absorber.*—The absorber shall be a 250-ml. gas washing bottle containing a gas washing tube having a fritted glass dispersion disk of medium porosity.

F–4c (1) e. *Gas absorber solution.*—The solution used to absorb the chlorine shall be approximately 0.1 normal with respect to sodium carbonate or hydroxide, and 0.1 normal with respect to sodium arsenite. The sodium arsenite may be readily obtained by dissolving the requisite amount of arsenic trioxide in a small quantity of sodium hydroxide. This arsenite solution shall be tested to ascertain the absence of halide.

F–4c (1) f. *Reagents.*—A standardized 0.1 normal solution of silver nitrate and 0.05 normal solution of potassium thiocyanate shall be employed to determine the inorganic chloride ion in the absorber solution. The ferric sulphate, nitrobenzene and concentrated nitric acid used in the determination shall be halogen free.

F–4c (2). *Procedure.*—Place approximately 50 ml. of the gas absorber solution in the absorber. Adjust the stopcock leading to the vacuum line to allow the minimum amount of air through the system necessary to support a flame 1 cm. high at the 1-mm. opening of the U-tube when fuel gas or hydrogen passing through the system is ignited at that point. Remove the U-tube from the apparatus, weigh[1] and introduce a 0.1 to 0.2 gm. sample of the DDT into the 9-mm.

[1] Weighings may be carried out by suspending the U-tube from the stirrup hook of an analytical balance by means of a wire loop.

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opening of the tube. Place the heating device, as shown on figure 1, over the U-tube which shall then be inserted into the apparatus. Adjust the flow of gas to produce a flame 1 cm. high at the 1-mm. capillary opening of the U-tube when the gas is ignited. Immediately place the chimney over the flame. Turn on the heating device and adjust so that the temperature rises slowly toward the sublimination point of the sample.[2] Soon after the sample has disappeared and the appearance of the flame indicates that it is all burned, stop the flow of gas, disassemble the apparatus, remove the chimney from the absorber, and rinse the gas washing tube, dispersion disk, and the inside of the absorber bottle cap carefully into the absorber bottle. Add a slight excess of the standard 0.1 normal silver nitrate solution to the solution in the absorber bottle from a burette. A sufficient excess of silver nitrate is indicated by the appearance of a yellow brown precipitate of silver arsenite in the alkaline solution. Add 2 ml. of concentrated nitric acid, 0.5 gm. of powdered ferric sulphate, and 3 ml. of nitrobenzene to the resultant solution for each 10 ml. of 0.1 normal silver nitrate solution used. Stir the mixture vigorously until the precipitate has coagulated, and then titrate the excess silver with the standard 0.05 normal thiocyanate solution.

F-4c (3). *Parr peroxide bomb method.*—If the Parr peroxide bomb method is used, the method shall be as prescribed in Navy Department Specification 52P63 with the following exceptions:

(a) The weight of the DDT sample taken shall be 0.2 gm.

(b) The weight of benzoic acid used shall be 0.4 gm.

(c) The amount of 0.1 normal silver nitrate solution added shall be sufficient to give a 1 to 3 ml. excess. Approximately 30 ml. will be needed for a 0.2 gm. sample of DDT.

F-4d. *Ash.*—Place a 2-gm. sample in a weighed crucible. Burn off or vaporize the DDT under a hood with good ventilation, over a low flame or from a sand bath. When the organic material has been substantially volatilized cool the crucible and add an excess of 10 percent sulfuric acid. The crucible shall be heated as above to dryness and then ignited in a muffle furnace for 1 hour at a temperature of 800° ±50° C. Transfer the crucible to a desiccator, cool and weigh.

F-4e. *Determination of pH.*—

F-4e (1). *pH by distillation.*—Place a 20-gm. sample of DDT and 200 ml. of carbon dioxide-free distilled water in a 500-ml. round bottom flask equipped with a mercury sealed mechanical stirrer. Heat the flask in an oil bath at a temperature between 140° and 160° C. with rapid stirring to prevent superheating, and distill the mixture through a well-cooled condenser at such a rate that 100 ml. of distillate shall be obtained in a period of not less than 30 minutes nor more than 1 hour. Collect exactly 100 ml. of distillate in a calibrated cone-shaped centrifuge tube similar to that illustrated by figure 2. The distillate is used to determine water-insoluble, steam volatile material (see par. F-4f) and chloral hydrate (see par. F-4g) as well as pH. Cool the residue in the flask to room temperature and filter. Determine the pH of the filtrate. Both distillate and filtrate must fall within the specified range.

F-4e (2). *pH by extraction (for Army purchases only).*—Transfer a 20 ±0.1-gm. sample of DDT to a 500 ml. separatory funnel and

[2] A trial run should be made to determine the volatilization characteristics of the DDT. If the vaporization rate is too great the flame may become smoky and soot may be deposited in the chimney. This must be avoided.

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dissolve in 100 ml. of benzene. Add 50 ml. of freshly-distilled, cooled carbon dioxide-free water, stopper, and shake the funnel and contents for 3 minutes. Allow the two phases to separate, and draw off the aqueous layer into a flask. Stopper this flask immediately. Repeat the aqueous extraction twice, using two successive 25-ml. portions of freshly-distilled, cooled carbon dioxide-free water. Determine the pH of this extract, using any suitable method. However, in the event of dispute, the results obtained with a calibrated pH electrometer shall govern. Reserve the remainder of this extract for the determination of chloral hydrate and water-soluble material.

F–4f. *Water insoluble, steam volatile material.*—Centrifuge the distillate (see par. F–4e (1)) until the volume of the water insoluble layer can be clearly read.

F–4g. *Chloral hydrate.*—The distillate (see par. F–4e (1)) or the aqueous extract (see par. F–4e (2)) may be used. Place 2 ml. of a sodium hydroxide solution (40 gm. in 100 ml. of solution) in a test tube, add 1 ml. of colorless pyridine, and 4 ml. of the distillate or extract. Similarly treat 4 ml. of a standard aqueous solution containing 0.05 mg. of chloral hydrate per ml. in another tube. Shake the two tubes and heat in a boiling water bath for 1 minute. The red color in the pyridine layer of the sample under test shall not be darker than that of the standard.

F–4h. *Water soluble material.*—Boil 50 ml. of the filtrate (see par. F–4e (1)) or of the aqueous extract (see par. F–4e (2)) down to a small volume, and dry the residue to constant weight at 105° C.

F–4i. *Cyclohexanone insoluble material.*—Place 71 ml. (67 gm.) of cyclohexanone in a calibrated cone-shaped centrifuge tube. (See fig. 2.) Add 33 ±0.1 gm. of DDT, stopper, and shake until the material is dissolved as completely as possible. Centrifuge for 10 minutes at 1,500 r. p. m., in an international size 1–5B centrifuge (Fisher No. 5–110 with head No. 240 and trunion carriers No. 5–285, or the equivalent) using a balanced system. The apparent volume of the separated solids is noted.

F–4j. *Monofluorotrichloromethane insoluble material.*—Place 5.0 gm. of DDT and 10.0 gm. of cyclohexanone in a calibrated centrifuge tube, as above. Stopper, shake until the material is dissolved as completely as possible. Dilute to 100 ml. with monofluorotrichloromethane. Stopper, shake, and centrifuge as described in paragraph F–4i. The solution in the tube should be clear and give not more than 0.01 ml. of sediment.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT.

G–1. *Packaging for domestic or overseas shipment.*—Unless otherwise specified, the subject commodity shall be furnished in 5-pound multiple friction top cans or 25-pound (5-gallon size) steel pails, as specified in the contract or order. (See par. H–2b.)

G–1a. Five pounds of the product shall be furnished in a round multiple friction top metal can made throughout of special coated manufacturers' terneplate and shall conform to the requirements of Joint Army-Navy Specification JAN–P–124, except that no exterior protective coating on the friction ring will be required.

G–1b. Twenty-five pounds of the product shall be furnished in a 5-gallon lug cover style steel pail conforming to the requirements of Joint Army-Navy Specification JAN–P–124.

G–2. *Packing.*—

G–2a. *For domestic shipment.*—

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G–2a (1). *Five-pound containers.*—Unless otherwise specified, six 5-pound cans, packaged as described in paragraph G–1a, shall be packed in regular slotted style corrugated or solid fiberboard boxes having a minimum Mullen test of 275 pounds. The cans shall be arranged within the box as follows: three in length, two in width, one in depth. In lieu of fiberboard boxes, six 5-pound cans shall be packed in cleated plywood boxes, nailed wood boxes, or wirebound boxes conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, and JAN–P–107, respectively. No strapping will be required.

G–2a (2). *Twenty-five pound containers.*—Twenty-five pound capacity (5-gallon steel pails) will require no further packing.

G–2b. *For overseas shipment.*—

G–2b (1). *Five-pound containers.*—Unless otherwise specified, six 5-pound cans shall be packed in cleated plywood boxes, nailed wood boxes, wirebound boxes, or in style RSC–SL fiberboard boxes, grade 1 or 2, conforming to the requirements of Joint Army-Navy Specifications JAN–P–105, JAN–P–106, JAN–P–107, and JAN–P–108, respectively. Each box shall be strapped in accordance with the requirements of the appendix of the applicable box specification. Closures of fiberboard boxes shall be effected by the application of a waterproof adhesive to the entire contact area of the flaps as specified in Joint Army-Navy Specification JAN–P–108.

G–2b (2). *Twenty-five pound containers.*—Twenty-five pound capacity (5-gallon) steel pails will require no further packing.

G–3. *Marking.*—

G–3a. *Army purchases.*—Marking shall be in accordance with the specification or contract under which DDT or the insecticide containing it is procured. Cans or containers shall be marked with one-print, black lithograph on can or container body or cover. Containers may be stenciled with weather-resistant ink.

G–3b. *Navy purchases.*—

G–3b (1). *Cans.*—Unless otherwise specified, cans, in addition to the labeling required in paragraph E–3, shall be clearly and indelibly marked with the following information completed:

MATERIAL _____
QUANTITY _____
CONTRACT NO. _____
CONTRACTOR _____
MANUFACTURER _____

G–3b (2). *Shipping containers and 25-pound capacity (5-gallon) steel pails.*—Unless otherwise specified, shipping containers and 5-gallon steel pails, in addition to the labeling required in paragraph E–3, shall be marked with the following information completed:

MATERIAL _____
QUANTITY _____
CONTRACTOR _____
CONTRACT NO. _____
GROSS WEIGHT _____

In addition to the foregoing, marking shall conform to the requirements of the latest issue of the Navy Shipment Marking Handbook.

H. NOTES.

H–1. Requests, requisitions, schedules, and contracts or orders should contain the title of the specification, the number, and date.

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H–2. Requests, requisitions, schedules, and contracts or orders should cover the following features:

(a) Grade of DDT required. (See par. B–1.)

(b) Size of containers required. (See par. G–1.)

(c) Whether the subject commodity is to be packed for domestic or overseas shipment. (See par. G–2.)

H–3. Since DDT is definitely toxic when ingested or when absorbed through the skin from its solutions, the following suggestions are made:

H–3a. *Plant hygiene.*—

H–3a (1). DDT should be manufactured in an isolated and closed system with proper exhaust ventilation throughout the plant.

H–3a (2). The vapors of chloral and chlorobenzene are known to be toxic and should be maintained at minimum levels. Maximum allowable concentrations in the working atmosphere have not yet been established for these substances, hence no limiting concentration in air can be designated.

H–3a (3). The best current practices in plant housekeeping should be carried out under supervision. All floors should be cleaned by flushing with water or by a vacuum cleaner.

H–3b. *Personal hygiene of plant employees.*—

H–3b (1). All workers should use coveralls and head coverings, laundered or dry-cleaned daily.

H–3b (2). Before each rest period workers should carefully wash their hands with the aid of a scrubbing brush to remove material from beneath the nails and cuticle. The face should also be carefully washed.

H–3b (3). All persons exposed to solutions of DDT in organic solvents should wear impervious gloves, sleeves, and aprons where necessary to prevent skin contact.

H–3b (4). All persons exposed to DDT dust in excess of 1.0 mg. per cubic meter of air should wear respirators of the type approved by the Bureau of Mines for protection against toxic dusts.

H–3b (5). Those persons concerned with operations where dust concentrations are less than 1.0 mg. per cubic meter should be instructed to wear a surgical gauze mask to minimize contact with face or mouth by fingers.

H–4. *Army purchases.*—For Army purchases, any change in the technical requirements of this specification or modifications at the time of purchase should be cleared with The Surgeon General, U. S. Army.

H–5. Copies of Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C.

H–6. Copies of Joint Army-Navy specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army and Federal Specifications Used by the War Department." Copies of this Index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department may obtain copies of Joint Army-Navy and United States Army specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–7. Copies of Joint Army-Navy specifications (required for Navy purchases) and Navy Department specifications may be obtained

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upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–8. Copies of this Joint Army-Navy specification for Army purchases may be obtained from:

Jersey City Quartermaster Depot, 84 Exchange Place, Jersey City 2, N. J.
Chicago Quartermaster Depot, 1819 West Pershing Road, Chicago 9, Ill.
California Quartermaster Depot, Clay and Fifteenth Streets, Oakland 4, Calif.
San Antonio Army Service Forces Depot, Grayson Street Station, San Antonio, Tex.
Jeffersonville Quartermaster Depot, Jeffersonville, Ind.
Kansas City Quartermaster Depot, Kansas City 1, Mo.
Philadelphia Quartermaster Depot, 2800 South Twentieth Street, Philadelphia 45, Pa.

H–9. Copies of this Joint Army-Navy specification for Navy purchases may be obtained as indicated in paragraph H–7.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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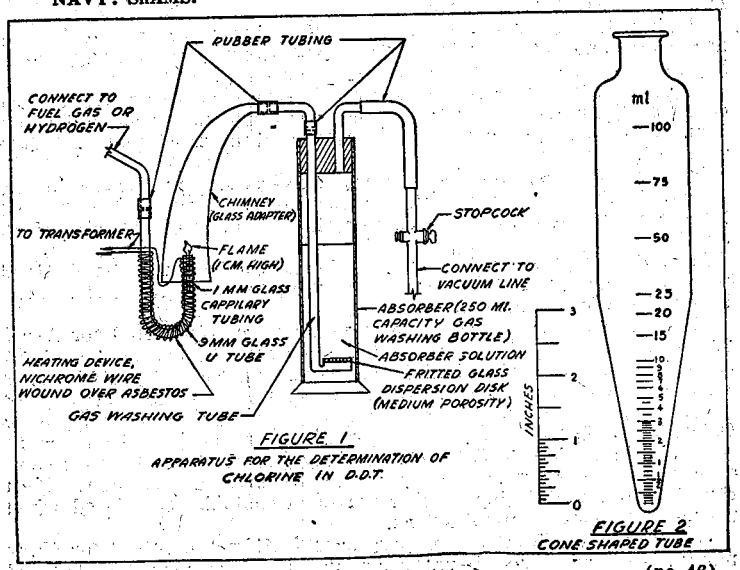


FIGURE 1
APPARATUS FOR THE DETERMINATION OF CHLORINE IN D.D.T.

FIGURE 2
CONE SHAPED TUBE

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